***Abstract*—This project focuses on the development and evaluation of a convolutional neural network (CNN) to classify bone marrow cell images derived from a curated and balanced subset of the Kaggle Bone Marrow Cell Classification dataset. A base CNN model is trained and evaluated using performance metrics such as accuracy, confusion matrices and per-class evaluation scores. The model is subsequently enhanced through three key improvements, data augmentation, batch normalization and dropout. These improvements significantly boosted generalization performance, as demonstrated through a comprehensive ablation study isolating the contribution of each component. Additionally, the project implements out-of-distribution (OOD) detection without retraining the model by leveraging softmax entropy scores. A histogram of the score distribution shows clear separation between in-distribution (ID) and out-of-distribution (OOD) samples, validating the approach. This work highlights the importance of model architecture improvements, training data enhancement and uncertainty estimation in developing robust image classifiers for medical applications.**

***Index Terms*—Bone Marrow Cell Classification, Model Optimization, Ablation Study, Out-of-Distribution (OOD) Detection.**

I. Introduction

Automated classification of bone marrow cells from microscopic images plays a vital role in assisting hematologists with the diagnosis and monitoring of blood-related disorders such as leukemia, anemia and infections. Traditionally, the task requires manual inspection of stained cell slides, which is time-consuming, subjective and requires extensive expertise. With the growing demand for scalable and accurate diagnostic support systems in clinical settings, there is a clear need for reliable automated methods to classify bone marrow cells.

Recent advances in deep learning, have shown remarkable success in computer vision tasks, including medical image classification. These deep learning models are capable of learning hierarchical representations directly from pixel-level data, reducing the need for handcrafted features and outperforming traditional approaches in many medical imaging applications.

To address the challenge of automating bone marrow cell classification from microscopic images, this project explores the use of convolutional neural networks (CNNs) trained on a balanced and preprocessed subset of the Kaggle Bone Marrow Cell Classification dataset consisting of only 8 classes of the 21 classes, namely, artefact (ART), blast (BLA), erythroblast (EBO), lymphocyte (LYT), band neutrophil (NGB), segmented neutrophil (NGS), plasma cell (PLM) and promyelocyte (PMO). The idea is to develop a simple CNN to establish the baseline performance and then follow with a few improvements to the model architecture, data and training to generate improved results.

The impact of these enhancements is assessed individually through a structured ablation study that isolates their contributions to performance. In the final phase, the project addresses the practical challenge of out-of-distribution (OOD) detection without modifying the trained network to distinguish unfamiliar inputs from known classes, simulating real-world uncertainty. This approach provides a complete pipeline, from classification to reliability assessment.

II. Method

A. Data Description, Pre-Processing and Loading

The classification task was addressed using a convolutional neural network (CNN) trained on a curated subset of the Kaggle Bone Marrow Cell Classification dataset. The dataset consists of $(56 \times 56 \times 3)$ JPEG images grouped into eight balanced classes, with 32,000 training and 24,000 test images.

The training data was loaded using the `ImageFolder` utility from the `PyTorch` library. To evaluate model performance during training, a stratified 90:10 split was applied to the training data using `StratifiedShuffleSplit` from the `sklearn` library, ensuring an even distribution of classes across both training and validation subsets. The test data, also evenly distributed across the same eight classes, was loaded in a similar manner.

All images were transformed into tensors and normalized using z-score normalization, with the mean and standard deviation computed from the training subset only, to prevent data leakage. These statistics were then applied uniformly to the training, validation and test datasets using `transforms.Normalize` from the `PyTorch` library.

Lastly, to facilitate efficient training and evaluation, `DataLoader` was used from the /textttPyTorch library to create iterable data batches of size 64 for the training, validation and test sets. Training loader set to shuffle the data to promote gradient stability and reduce training bias while the validation and test loader, in contrast, processed data in a fixed order with no shuffling, to ensure reproducibility during evaluation.

B. Base Model

The base convolutional neural network (CNN) model is designed to process input images of shape $(56 \times 56 \times 3)$. The architecture begins with a convolutional layer that applies 32 filters of size 3×3, with padding set to 1 to preserve spatial dimensions. This is followed by a ReLU activation function and a 2×2 max-pooling layer, reducing the feature map size to $(28 \times 28 \times 32)$.

The second convolutional layer uses 64 filters of size 3×3, again with padding 1, followed by ReLU activation and another 2×2 max-pooling operation, resulting in a feature map of size $(14 \times 14 \times 64)$. This is succeeded by a third convolutional layer, also with 64 filters and ReLU activation, followed by another max-pooling layer that reduces the size to $(7 \times 7 \times 64)$.

A fourth convolutional layer is then applied, consisting of 128 filters with the same 3×3 kernel and padding 1. ReLU activation is used again and a final 2×2 max-pooling operation reduces the spatial dimensions to $(3 \times 3 \times 128)$. The resulting tensor is flattened to a one-dimensional vector of size 1152.

This flattened representation is passed through a fully connected dense layer with 256 neurons and ReLU activation, capturing high-level features from the learned spatial representations. Finally, the output layer consists of 8 neurons corresponding to the number of classes, producing the raw logits.

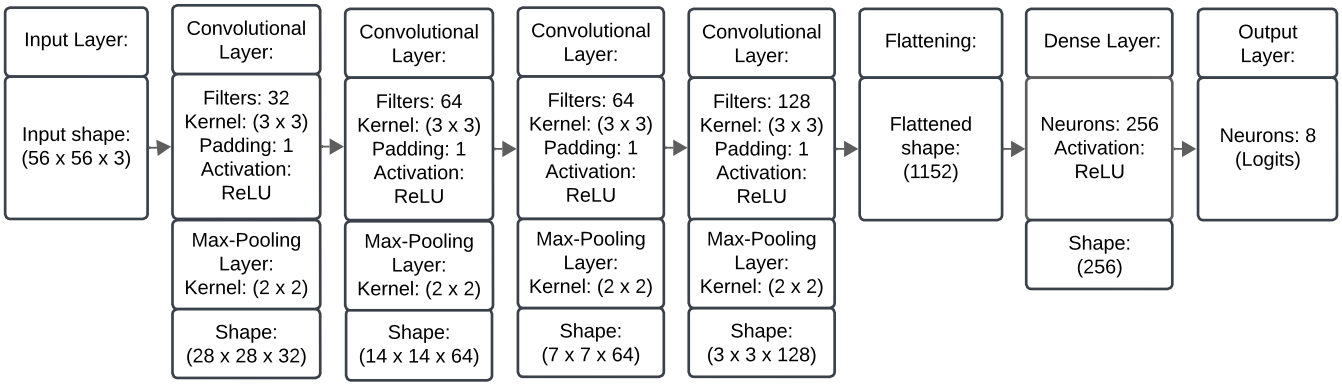


Fig. 1. Base / Simple CNN architecture used to generate the baseline results.

The visualization of the base model architecture can be seen in Fig. 1.

C. Improvements

To address challenges in training deep leaning models, three modifications / improvements were made namely, data augmentation, batch normalization and dropout. These were introduced to improve robustness to input variation, stabilize optimization and reduce overfitting.

Data Augmentation: A combination of random horizontal flips (with a probability of 0.5) and random rotations in the range $[-10°, +10°]$ was applied to the training images. These augmentations artificially increase dataset diversity without requiring additional labeled data. This combats overfitting by exposing the model to a wider range of plausible variations, helping it learn more generalizable features rather than memorizing training patterns.

Batch Normalization: Batch normalization layers were inserted before each ReLU activation in the convolutional blocks. This technique normalizes the outputs of each mini-batch to have zero mean and unit variance, reducing internal covariate shift. This stabilization of activation distributions across layers accelerates convergence, making the training process more efficient and reliable.

Dropout: A dropout layer with a rate of 0.5 was added after the first fully connected layer. Dropout randomly disables a subset of neurons during each training iteration, which prevents the network from becoming overly reliant on specific pathways. This forces the model to learn distributed representations, reducing the likelihood of overfitting to training data and enhancing generalization to unseen samples.

Together, these improvements address distinct bottlenecks in the base model and their individual contributions are analyzed further in the ablation study. The resulting improved model architecture, incorporating all three enhancements, is illustrated in Figure 2.

D. Out-of-Distribution Detection

To extend the classifier's reliability in real-world settings, an out-of-distribution (OOD) detection mechanism was incorporated without retraining or modifying the network. The approach leverages the softmax output of the trained classifier to estimate the uncertainty of predictions. Specifically, a scalar score is computed for each sample based on the entropy of the softmax probability distribution. This entropy score captures how confidently the model makes a prediction, in-distribution (ID) samples typically yield low entropy, while OOD samples lead to higher entropy due to more uniformly distributed softmax outputs. Given the softmax output vector

$$\mathbf{p} = [p_1, p_2, \ldots, p_K],$$

where p_i represents the predicted probability for class i among K total classes, the entropy score S is computed as:

$$S = -\sum_{i=1}^{K} p_i \log(p_i + \epsilon),$$

where ϵ is a small constant (e.g., 10^{-10}) added for numerical stability.

This method enables ID-OOD detection directly from the model's output layer without requiring architectural changes or additional training data.

III. Experimental Evaluation

A. Training Setup

All models, including the base network, the improved version and the ablation variants, were trained using the Adam optimizer with a fixed learning rate of 0.001. The loss function used was standard unweighted cross-entropy loss, as the dataset was balanced across the eight classes. Each model was trained for a total of 15 epochs. While no explicit early stopping criterion was used, the model weights from the epoch yielding the highest validation accuracy were retained, effectively serving as an early stopping mechanism. This consistent training configuration ensures a fair and controlled comparison across all model variants.

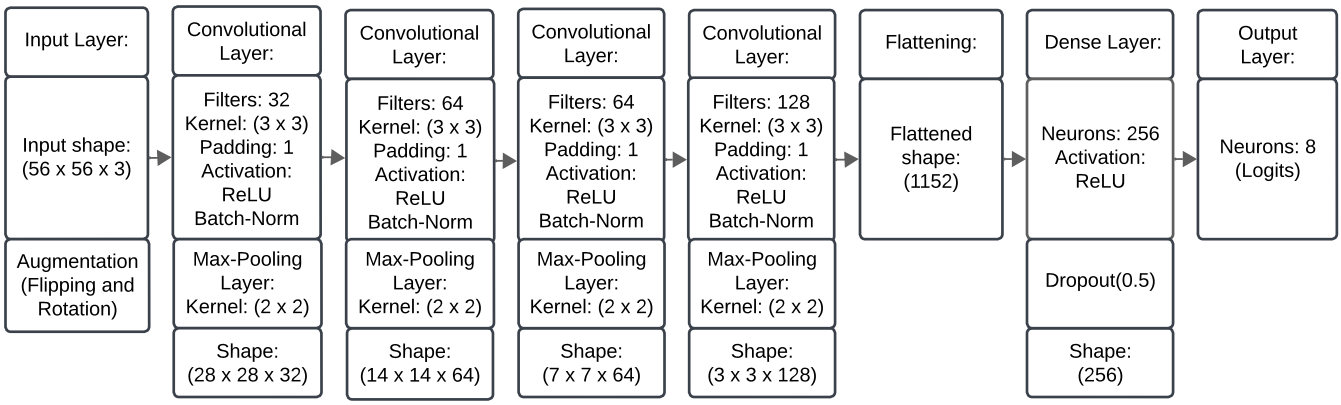


Fig. 2. Improved CNN architecture used to enhance baseline model performance.

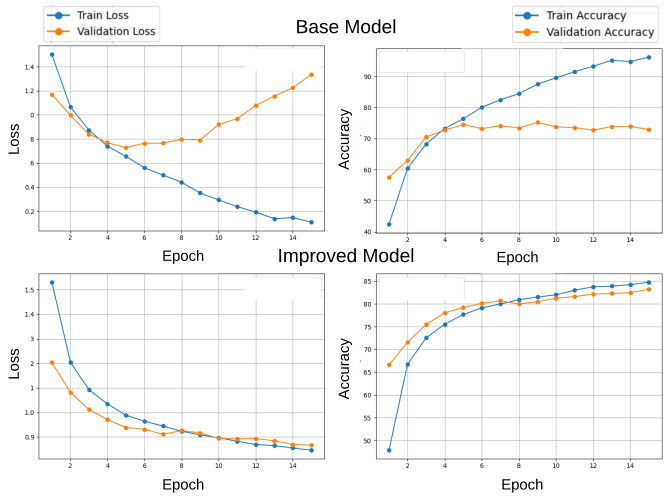


Fig. 3. Training and validation loss and accuracy for the base and improved models. The improved model shows better convergence and overfitting.

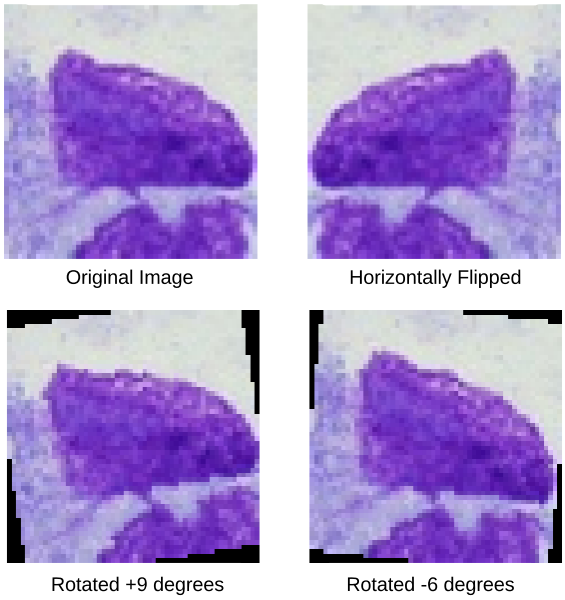


Fig. 4. Examples of data augmentation applied during training. The original image (top left) is randomly flipped and rotated to simulate spatial variation.

B. Effect of Improvements

The effect of the architectural improvements were analyzed by comparing training and validation dynamics between the base model and the improved model.

Figure 3 shows the training and validation loss and accuracy over 15 epochs for both models. The base model exhibited more pronounced divergence between training and validation curves, a sign of overfitting and unstable training. In contrast, the improved model showed smoother and more consistent convergence. This improvement is attributed to the inclusion of batch normalization, which stabilizes learning and dropout, which regularizes the network by reducing overfitting.

To demonstrate the effect of data augmentation, Figure 4 displays examples of transformations applied to a single training image. These include a horizontal flip and small-angle rotations of $\pm 9^\circ$ and -6°, respectively. Such augmentations increase the diversity of the training dataset by simulating natural variations in image orientation. This helps the model become more robust to positional shifts and improves its generalization to unseen data without requiring additional labeled samples.

C. Ablation Impacts

An ablation study was conducted to evaluate the impact of each enhancement on model performance. Three models were trained, each with one component, data augmentation, batch normalization or dropout, removed, while the others remained unchanged. This approach enabled a clear assessment of each feature's contribution to training stability and overall learning generalization, clearly highlighting their individual effectiveness.

D. Ablation Study 1: No Augmentation

In this variant, the model retained batch normalization and dropout, but no data augmentation was applied during training. The corresponding training and validation loss and accuracy curves are shown in Figure 5.

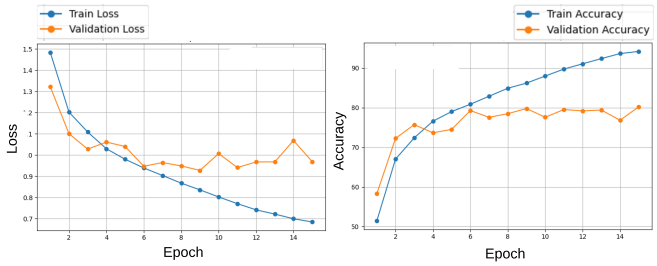


Fig. 5. Training and validation performance for the model without data augmentation.

Compared to the improved model, this version exhibited a more noticeable gap between training and validation, loss and accuracy curves. The lack of variability in training data caused the model to generalize less effectively. These results demonstrate that data augmentation plays a key role in increasing robustness by exposing the model to a wider range of plausible input transformations.

E. Ablation Study 2: No Batch Normalization

In this variant, the model retained data augmentation and dropout, but batch normalization was removed from all the convolutional blocks. The training and validation performance for this configuration is shown in Figure 6.

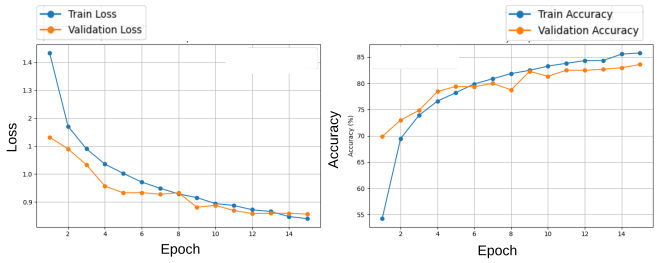


Fig. 6. Training and validation performance for the model without batch normalization.

The exclusion of batch normalization resulted in slightly noisier training and validation curves. Unlike the version without data augmentation, this model did not exhibit significant divergence between training and validation, loss and accuracy. However, the learning dynamics were less smooth and more prone to fluctuations across epochs. This behavior reflects the typical role of batch normalization, not as a primary regularizer, but as a stabilizer that smooths optimization by reducing internal covariance shift. While its impact on final performance is usually less pronounced compared to the other improvements, its contribution to training stability remains beneficial.

F. Ablation Study 3: No Dropout

In this variant, the model retained data augmentation and batch normalization, but the dropout layer in the fully connected head was removed. The training and validation curves for this configuration are shown in Figure 7.

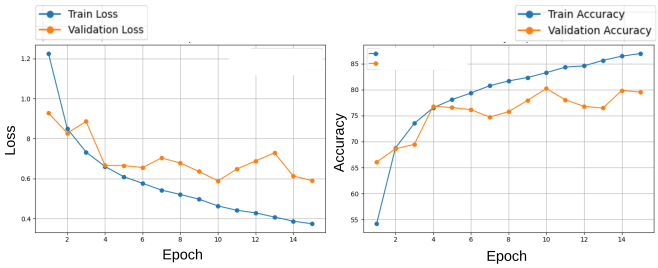


Fig. 7. Training and validation performance for the model without dropout.

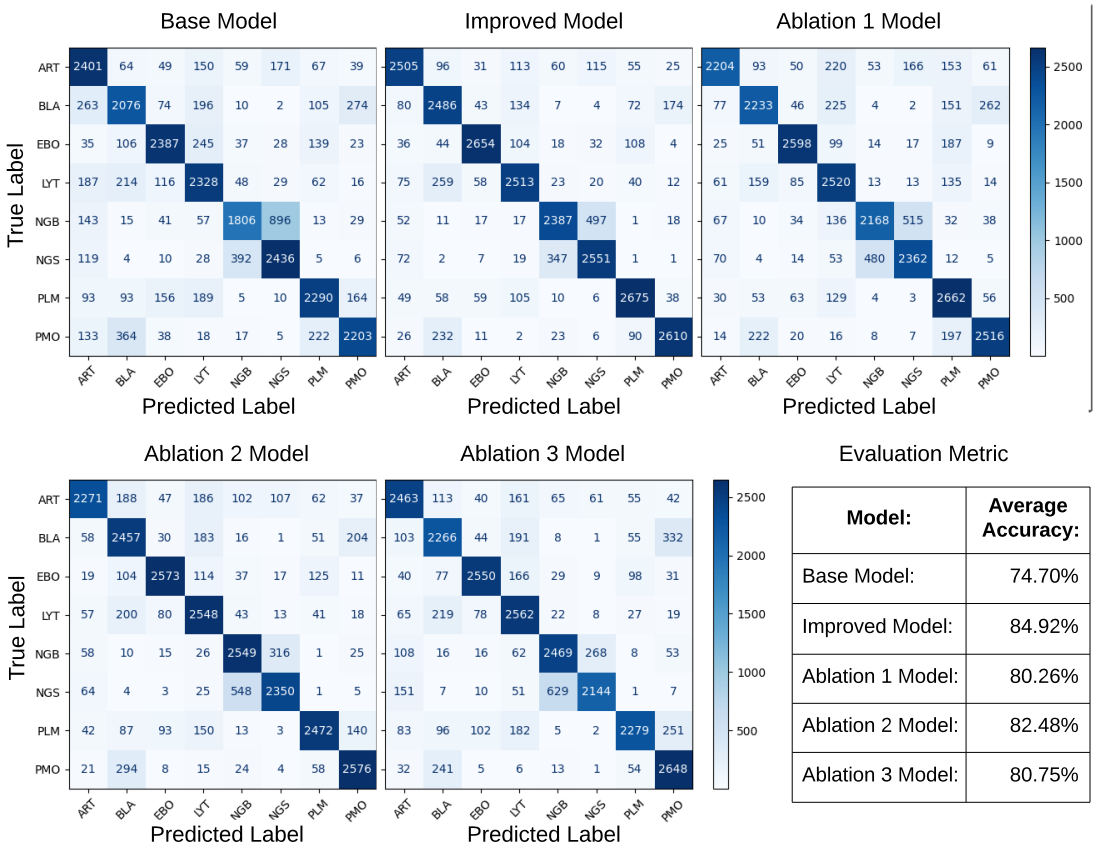
The exclusion of dropout led to a visibly sharper rise in training accuracy, indicating that the model was able to fit the training data more aggressively. However, this came at the cost of a growing gap between training and validation accuracy over the epochs, reflecting classic symptoms of overfitting. These effects underscore dropout's effectiveness as a regularization technique that improves generalization. Without it, the model learns more specific representations of the training set, which do not transfer as well to unseen data.

G. Evaluation Results

The final classification performance of the base model, the fully improved model and each of the three ablation variants is presented in Figure 8, which shows the confusion matrices and average test accuracy for all five models. All models were trained and evaluated on the same data splits to ensure consistency and fairness in comparison.

As seen in the results, the base model achieved an average test accuracy of 74.70%. Incorporating all three enhancements in the improved model led to a significant increase in accuracy to 84.92%, demonstrating the cumulative benefit of the selected improvements.

Among the ablation variants, the absence of either data augmentation or dropout led to noticeable declines in performance. Specifically, the model without augmentation (ablation 1), achieved an accuracy of 80.26%, while the model without dropout (ablation 3), reached 80.75% highlighting the comparatively stronger influence of both the techniques in helping the model learn generalizable features. On the other hand, removing batch normalization (ablation 2), had a more subtle effect, with the model achieving 82.48% accuracy. While batch normalization contributed to smoother training and more stable learning curves, its absence did not degrade the final performance as significantly as the other two components. These results emphasize the stronger influence of data augmentation and dropout, while batch normalization, despite its comparatively smaller effect, still provided meaningful performance gains.



Evaluation Metric

Model:	Average Accuracy:
Base Model:	74.70%
Improved Model:	84.92%
Ablation 1 Model:	80.26%
Ablation 2 Model:	82.48%
Ablation 3 Model:	80.75%

Fig. 8. Confusion matrices and average test accuracy for the base model, improved model and each ablation variant.

In addition to differences in average accuracy, the confusion matrices exhibit a consistent structural pattern across all models. Certain class pairs are more prone to misclassification, with one of the most prominent examples being the difficulty in distinguishing between band and segmented neutrophils, likely due to their strong visual similarity. While the overall structure of misclassifications remains similar, higher-performing models naturally produce sharper and more confident diagonals in the confusion matrix, reflecting fewer errors and improved class separability.

The ablation study confirms that the proposed improvements achieved their intended effects. Data augmentation and dropout effectively reduced overfitting, as evidenced by the comparatively lower deviation, compared to the base model, between training and validation performance in the presence of either of the two improvements and a noticeable decline in accuracy when either was removed. Batch normalization, while contributing less to the final accuracy compared to the other improvements, clearly stabilized the training process and provided a quantifiable benefit. Although the drop in average accuracy without it was relatively modest, it was nonetheless observable, thus confirming that batch normalization does indeed play a meaningful role. Not only did each technique yield improvements, but they also did so in alignment with what was expected, that is augmentation and dropout had a more significant impact while batch normalization offered a smaller yet somewhat significant benefit. These results validate the initial motivation behind incorporating each of the three enhancements confirming their individual and combined importance.

H. Qualitative failure analysis

To further understand the behavior of the improved model, a subset of correctly and incorrectly classified test images is shown in Figure 9. Each image is bordered in green if the model's prediction matched the actual (true) label and in red otherwise. The class names shown above each image indicate the predicted and actual (true) labels.

The images in this dataset are relatively low-resolution, with a size of only 56×56 pixels. This limited spatial resolution often results in visual ambiguity, especially for classes with subtle structural differences. For instance, as seen above, the model occasionally misclassifies band neutrophils (NGS) as segmented neutrophils (NGB) and vice versa as these two classes are assumed to be structurally very similar and even challenging to distinguish manually at such resolution.

Despite these challenges, the model demonstrates consistent and accurate predictions in many cases, indicating that it has learned to extract relevant features. Nonetheless, the misclassi-

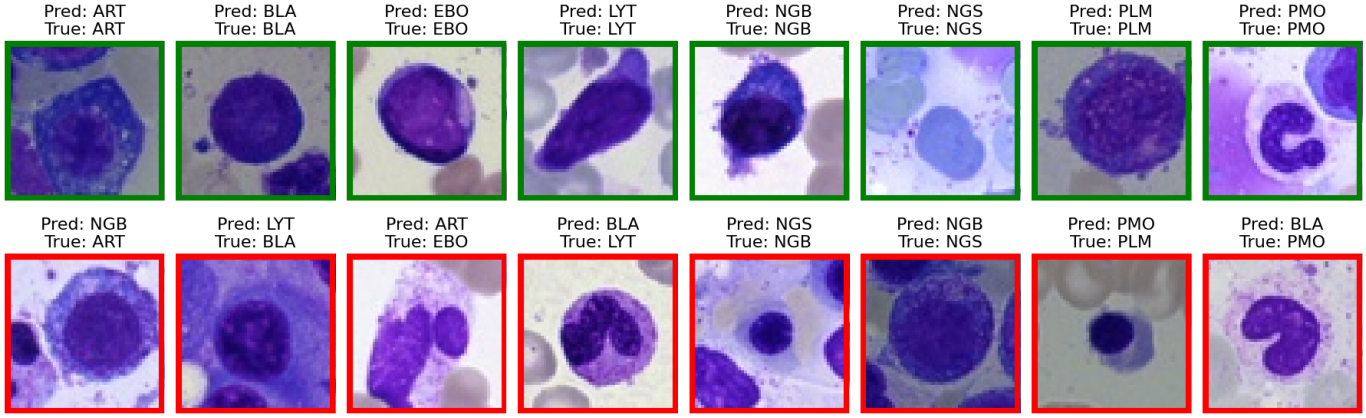


Fig. 9. Sample predictions from the improved model. Green borders indicate correct classifications, red borders indicate misclassifications.

fications reveal the inherent limitations of both resolution and class structural similarity within the dataset.

I. *Out-of-Distribution Analysis*

Figure 10 shows a histogram of the computed scores, with bars colored according to their assumed category. Samples with low entropy are considered to be in-distribution (ID) data, as they indicate confident predictions with one dominant softmax output. In contrast, samples with higher entropy likely correspond to out-of-distribution (OOD) data, where the model's predictions are uncertain and the probability mass is distributed across multiple classes.

The resulting distribution is clearly bimodal. The leftmost cluster (in green) shows a sharp peak near zero, corresponding to ID samples. The rightmost spread (in red) exhibits a broader, flatter distribution indicative of the OOD samples.

This separation confirms that softmax entropy is a viable metric for distinguishing OOD from ID data.

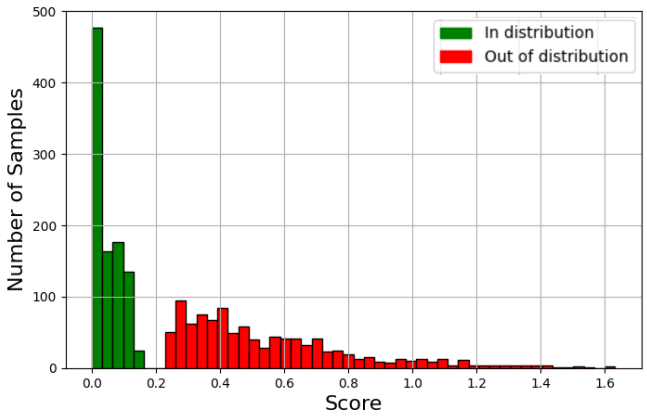


Fig. 10. Histogram of softmax entropy scores computed on the mixed ID-OOD dataset. Lower scores indicate confident predictions, characteristic of in-distribution samples, while higher scores reflect uncertainty, characteristic of out-of-distribution samples.

IV. Conclusion

This project presented a convolutional neural network based approach for classifying bone marrow cell images into eight distinct categories using a balanced dataset derived from the Kaggle Bone Marrow Cell Classification dataset. Starting with a simple baseline architecture, the model was systematically improved through the integration of three key techniques, data augmentation, dropout and batch normalization. Each of these enhancements addressed specific challenges and together they led to substantial increase in model efficiency.

Experimental evaluations demonstrated that the improved model significantly outperformed the baseline, achieving a notable increase in test accuracy and an ablation study further highlighted the individual impact of each improvement, with augmentation and dropout providing more significant improvements, while although batch normalization contributed comparatively less but still provided a marginal improvement.

Finally, the model was tested on out-of-distribution data using softmax entropy scores. This helped separate in-distribution and out-of-distribution samples, showing that the model could detect unfamiliar inputs.